

Backend Developer Home Assignment

Objective

Develop a mini backend application for a cryptocurrency dashboard.

The application should expose an API that allows users to retrieve cryptocurrency data, manage a personal watchlist, and access historical price information.

There is no requirement to build a user interface. The backend should provide the data that could later be displayed in a dashboard.

Overview

The application should allow users to:

1. View a dashboard-style list of current cryptocurrency rates.
 2. View detailed information for a specific cryptocurrency.
 3. Add cryptocurrencies to a personal watchlist.
 4. Persist the watchlist per authenticated user.
 5. Retrieve historical price data for selected cryptocurrencies.
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Backend Requirements

The backend should include:

1. A clear API structure.
2. User authentication.
3. Persistent data storage.
4. Integration with an external cryptocurrency data provider.
5. Proper error handling and validation.
6. Clean, readable, and maintainable code.

You may choose any backend language, framework, database, ORM, query builder, or architectural style, as long as the project is well structured and the implementation is clear.

Required Features

API Endpoints

The API should support at least the following capabilities:

1. Get a list of cryptocurrencies.
2. Get details of a single cryptocurrency.

3. Add a cryptocurrency to the authenticated user's watchlist.
4. Retrieve the authenticated user's watchlist.
5. Refresh cryptocurrency data from an external provider.
6. Retrieve historical price data for a selected cryptocurrency.

Suggested routes:

```
GET /api/cryptocurrencies
GET /api/cryptocurrencies/:id
POST /api/watchlist
GET /api/watchlist
POST /api/cryptocurrencies/refresh
GET /api/cryptocurrencies/:id/history
```

Additional routes may be added if they improve the platform functionality.

Data Model

The system should store cryptocurrency-related data with fields such as:

```
name
symbol
externalProviderId
currentPrice
marketCap
priceChangePercentage
watchlistStatus / relation to user watchlist
lastUpdatedAt
```

You may add any additional fields that are useful for the application.

External API Integration

Use a known cryptocurrency data provider to fetch:

1. Current cryptocurrency rates.
2. Details for a specific cryptocurrency.
3. Historical price data.

The system should include a way to refresh coin data from the external provider.

Deployment - Bonus

Deploy the backend application to any cloud or free hosting service.

The README should include deployment instructions.

Bonus points for adding a CI/CD pipeline using a common automation tool.

Optional Bonuses

1. API documentation using Swagger/OpenAPI or another documentation tool.
 2. Integration with an AI service that receives historical price data and returns a short analysis of a selected coin.
 3. CI/CD pipeline for automatic deployment.
 4. Additional useful endpoints or features that improve the platform.
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Deliverables

1. Public GitHub repository with the source code.
 2. README file with:
 3. Project overview
 4. Setup instructions
 5. Environment variables
 6. How to run locally
 7. API usage instructions
 8. Deployment instructions
 9. Working deployed application URL, if deployed.
 10. API documentation:
 11. Postman collection is required
 12. Swagger/OpenAPI is a bonus
 13. A short video, up to 5 minutes, explaining:
 14. What was built
 15. Main features
 16. API demonstration
 17. Important technical decisions
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Requirements

1. All API calls should work through Postman or a similar API client.
2. The GitHub repository must be publicly accessible.
3. The README should clearly explain how to run and test the project.
4. The code should be readable, structured, and follow best practices.

5. The application should handle errors and edge cases gracefully.
 6. Configuration values such as API keys, secrets, and database connection strings must be managed using environment variables.
 7. The solution should be completed within 4 days.
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Submission

Submit the assignment via the personal drive folder, including:

1. GitHub repository link.

Good luck.